

Non-Hodgkin Lymphoma — Subtypes, Viral Drivers & Staging



1. NHL incidence (~4% / non-Hodgkin)

WHAT YOU SEE

4% banner, NO-HODGWIG

WHAT IT ENCODES

NHL is far more common than Hodgkin and rises with age and immunosuppression. The 'census' frames the relative frequencies of subtypes. ~85-90% are B-cell origin.

BOARD TRAP

Reed-Sternberg cells and bimodal age + contiguous nodal spread are Hodgkin; NHL spreads non-contiguously and is more often extranodal.

2. DLBCL (most common, 31%)

WHAT YOU SEE

DLBCL 31% on board

WHAT IT ENCODES

Diffuse large B-cell lymphoma is the most common NHL — aggressive but curable with R-CHOP. Hans algorithm splits GCB vs ABC (non-GCB); double-hit (MYC+BCL2/BCL6) is high-grade with worse outcome.

BOARD TRAP

DLBCL is aggressive yet potentially curative — don't treat it as incurable like indolent FL; intent is cure.

3. Follicular lymphoma (22%, indolent)

WHAT YOU SEE

FL 22%

WHAT IT ENCODES

Follicular lymphoma is the prototypical indolent B-NHL with t(14;18)/BCL2 overexpression. Waxing-waning adenopathy; incurable but long survival. Watchful waiting if low burden; FLIPI for prognosis.

BOARD TRAP

Rapid growth/LDH spike in known FL suggests transformation to DLBCL (Richter-type) — rebiopsy.

4. B-cell predominance (~90%)

WHAT YOU SEE

Big blue B 90%

WHAT IT ENCODES

~85-90% of NHL is B-cell. CD20 positivity makes them targetable with rituximab — the backbone of immunochemotherapy. T/NK-cell lymphomas (~10%) are generally more aggressive and CD20-negative.

BOARD TRAP

Rituximab works only on CD20+ B-cells — useless for T-cell or NK/T lymphomas.

5. 5-yr survival ~74%

WHAT YOU SEE

SURVIVAL 74% 5-YR

WHAT IT ENCODES

Aggregate NHL 5-year survival is favorable but highly subtype-dependent: indolent lymphomas have long median survival; aggressive ones are curable; some T-cell subtypes do poorly.

BOARD TRAP

Don't quote a single survival number — prognosis hinges on subtype, IPI/FLIPI, and cell of origin.

6. EBV-driven lymphomas

WHAT YOU SEE

WANTED EBV bottle

WHAT IT ENCODES

EBV drives Burkitt (endemic), post-transplant lymphoproliferative disease, primary CNS lymphoma in HIV, and some Hodgkin and NK/T-cell lymphomas. Reactivation in immunosuppression is key.

BOARD TRAP

EBV is a passenger/driver in many lymphomas — it's not specific to one; pair with the clinical context (transplant, HIV, jaw mass).

7. HTLV-1 → ATLL

WHAT YOU SEE

WANTED HTLV-1

WHAT IT ENCODES

HTLV-1 causes adult T-cell leukemia/lymphoma — endemic Japan/Caribbean. Presents with hypercalcemia, lytic bone lesions, skin lesions, and 'flower cells.' Aggressive, poor prognosis.

BOARD TRAP

Hypercalcemia + lymphoma in a Caribbean/Japanese patient = ATLL (HTLV-1), not multiple myeloma.

8. HIV-associated lymphoma

WHAT YOU SEE

WANTED HIV

WHAT IT ENCODES

HIV raises risk of aggressive B-NHL — DLBCL, Burkitt, primary CNS lymphoma (EBV+), and primary effusion lymphoma (HHV-8). Lower CD4 correlates with CNS/EBV-driven types.

BOARD TRAP

A ring-enhancing CNS lesion in HIV that's EBV+ on CSF PCR favors PCNSL over toxoplasmosis.

9. HCV → marginal zone

WHAT YOU SEE

WANTED HCV

WHAT IT ENCODES

Chronic hepatitis C is linked to splenic/marginal zone lymphoma and mixed cryoglobulinemia. Antiviral therapy can induce lymphoma regression in some HCV-associated indolent cases.

BOARD TRAP

In HCV-driven indolent marginal zone lymphoma, treat the virus first — DAAs may shrink the lymphoma without chemo.

10. H. pylori → gastric MALT

WHAT YOU SEE

WANTED H. pylori stomach

WHAT IT ENCODES

H. pylori drives gastric MALT (extranodal marginal zone) lymphoma. Early-stage disease often regresses with eradication antibiotics alone. t(11;18) predicts antibiotic resistance.

BOARD TRAP

t(11;18)-positive gastric MALT won't respond to H. pylori eradication — needs radiation/systemic therapy.

11. HHV-8 → primary effusion / Castleman

WHAT YOU SEE

WANTED HHV-8

WHAT IT ENCODES

HHV-8 (KSHV) causes primary effusion lymphoma (body-cavity effusions, no mass; often HIV+) and multicentric Castleman disease, plus Kaposi sarcoma.

BOARD TRAP

Malignant effusions without a discrete mass in HIV = primary effusion lymphoma (HHV-8), not metastatic carcinoma.

12. Burkitt lymphoma

WHAT YOU SEE

WANTED BL

WHAT IT ENCODES

Burkitt is the most aggressive B-NHL — MYC t(8;14), 'starry-sky' histology, near-100% Ki-67. Endemic (jaw, EBV), sporadic (abdominal), HIV-associated. Cured with intensive dose-dense chemo + intense tumor lysis prophylaxis.

BOARD TRAP

Burkitt has the highest tumor lysis risk — pre-treat with rasburicase/hydration; standard R-CHOP is insufficient.

13. NK/T-cell lymphoma

WHAT YOU SEE

WANTED NK/T

WHAT IT ENCODES

Extranodal NK/T-cell lymphoma, nasal type — EBV-driven, midline destructive nasal/palate lesions, CD20-negative. Treated with asparaginase-based regimens + radiation, not R-CHOP.

BOARD TRAP

Rituximab fails here (CD20-negative); a destructive midline nasal mass in an Asian/Latin American patient = NK/T.

14. Ann Arbor staging

WHAT YOU SEE

Ann Arbor body map I-IV, thermometer

WHAT IT ENCODES

Stages I-IV by nodal regions and sites; 'B symptoms' (fever, drenching sweats, >10% weight loss) add 'B'. Stage IV = extranodal/marrow/liver. Staging guides curative vs palliative intent.

BOARD TRAP

In NHL, IPI/FLIPI predict outcome better than stage alone — stage is necessary but not sufficient.

15. IPI / FLIPI scores

WHAT YOU SEE

IPI (APLES) & FLIPI (ANHLS) cards

WHAT IT ENCODES

IPI for aggressive NHL: Age >60, Performance status, LDH, Extranodal sites, Stage. FLIPI for follicular: Age, Nodal areas, Hb, LDH, Stage. More risk factors → worse survival.

BOARD TRAP

Use FLIPI for follicular and IPI for DLBCL — applying the wrong index misclassifies prognosis.

16. Mandatory HBV screen

WHAT YOU SEE

HBV SCREEN mandatory checkbox

WHAT IT ENCODES

Screen for hepatitis B (HBsAg, anti-HBc) before rituximab — anti-CD20 therapy can reactivate HBV causing fulminant hepatitis. Give antiviral prophylaxis (entecavir/tenofovir) in carriers.

BOARD TRAP

Skipping HBV screening before rituximab risks fatal HBV reactivation — it's mandatory, not optional.